TechCrush Capstone Project

FTP vs SFTP Security Demonstration

Group 6

Group No. 6

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Outline

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Aim and Objective

- Demonstrate the security risk of transferring data with unsecured FTP.
- Show that FTP exposes commands, credentials and file content in readable plaintext.
- Repeat the same transfer using SFTP over SSH on port 22.
- Prove from packet capture evidence that SFTP keeps credentials and file payload encrypted.
- Compare both protocols and explain why SFTP is preferred for secure file transfer.

Core lesson: FTP moves files, but SFTP protects the transfer.

Concept of the Project

The project uses one controlled file-transfer scenario and compares what a packet capture can reveal under two protocols.

- FTP test: transfer secret.txt over vsftpd and inspect the captured traffic.
- SFTP test: transfer the same file through OpenSSH and inspect SSH traffic on port 22.
- Comparison point: what is readable to an observer with Wireshark access.
- Expected result: FTP reveals the username, password and file content; SFTP only reveals encrypted SSHv2 packets.

Methodology

- Set up vsftpd and create the sample file secret.txt.
- Connect to the FTP server at 192.168.1.10 using testuser and pass1234.
- Capture FTP traffic with the filter ftp || ftp-data and inspect USER, PASS, RETR and FTP-DATA.
- Start the OpenSSH service and connect with sftp testuser@192.168.1.10.
- Capture SFTP traffic with tcp.port == 22 and inspect SSHv2 encrypted packets.
- Compare both captures using confidentiality, credential safety, port usage and real-world risk.

Tools Used and Lab Environment

- Kali terminal: setup and file-transfer commands.
- vsftpd 3.0.5: FTP server running on port 21.
- OpenSSH server: SFTP/SSH service running on port 22.
- Wireshark 4.2.0: captured and inspected FTP and SFTP traffic.
- Sample file: secret.txt, transferred through both protocols.
- Filters: ftp || ftp-data for FTP; tcp.port == 22 for SFTP.

FTP Setup and File Transfer

The FTP service is started on port 21. The client logs in as testuser and retrieves secret.txt successfully.

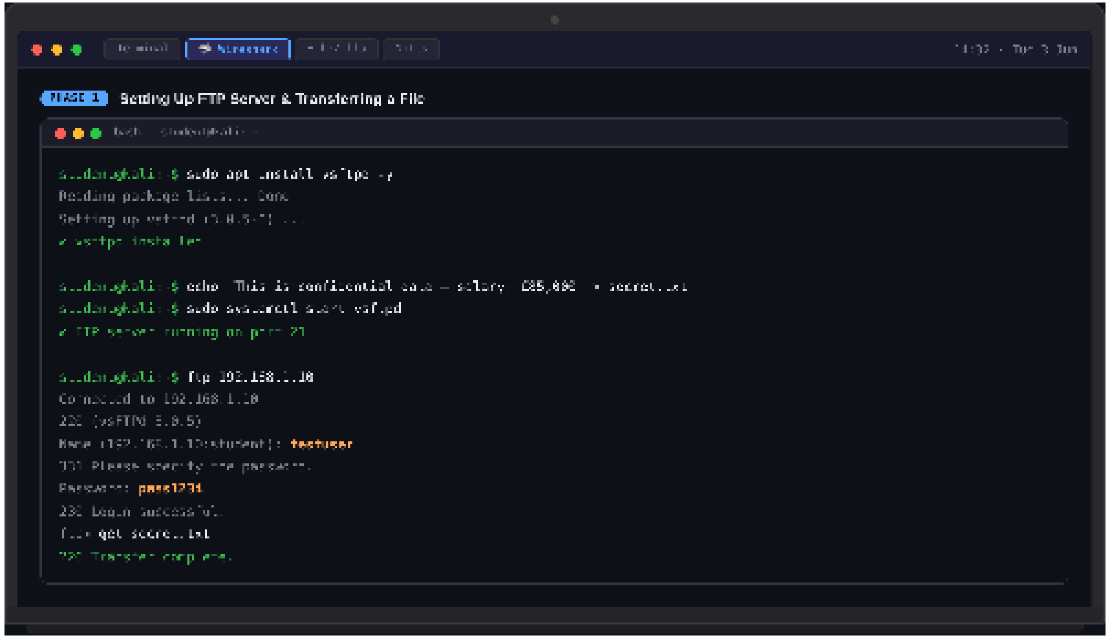

Figure 1. FTP setup and transfer: vsftpd runs on port 21, login succeeds and 226 Transfer complete confirms the file transfer.

FTP Packet Capture Evidence

Wireshark shows FTP commands and payload in plaintext. USER, PASS and the file content are visible from the capture.

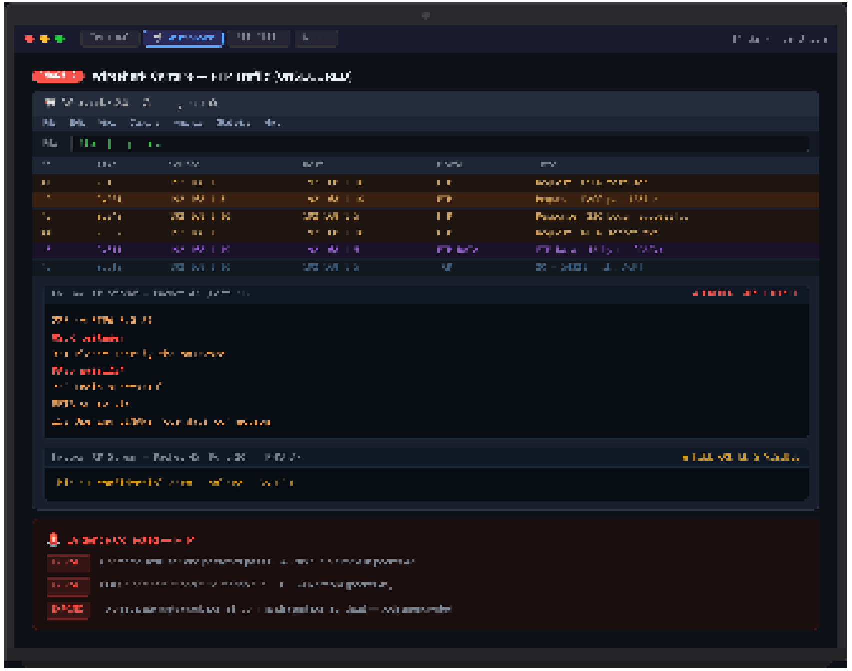

Figure 2. FTP capture: packet 42 exposes PASS pass1234 and packet 45 exposes the transferred file content.

SFTP Setup and File Transfer

OpenSSH runs on port 22. The SFTP client connects to the same server and retrieves the same file.

Figure 3. SFTP setup and transfer: OpenSSH is running on port 22 and secret.txt is retrieved successfully.

SFTP Packet Capture Evidence

The SFTP capture shows SSHv2 traffic on port 22 and encrypted AES-256-CTR packets. No readable USER, PASS or file text appears.

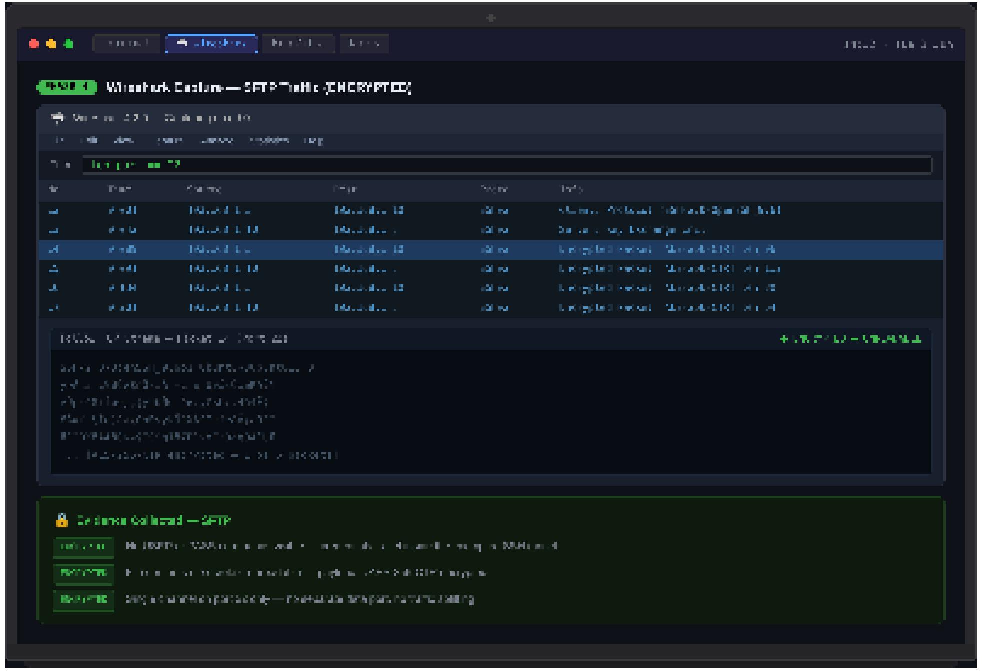

Figure 4. SFTP capture: SSHv2 packets are visible, but credentials and file payload are encrypted and unreadable.

Comparative Security Results

FTP Result

• Encryption: no protection; commands, credentials and file content are plaintext.
• Credential safety: USER testuser and PASS pass1234 are visible.
• File confidentiality: secret.txt content is readable in FTP-DATA.
• Ports/channels: command traffic and FTP-DATA are separate.
• Risk: sniffing can expose login details and sensitive files.

SFTP Result

• Encryption: protected through SSH; capture shows encrypted SSHv2 packets.
• Credential safety: no readable USER or PASS commands.
• File confidentiality: payload appears as encrypted data.
• Ports/channels: single SSH channel on port 22.
• Risk: capture confirms a session, but not the useful content.

Real-World Implications

- FTP is unsafe on shared, public or untrusted networks because a packet capture can reveal credentials and sensitive files.
- Exposed transfers can affect employee records, customer data, financial files, contracts and internal business information.
- SFTP is preferred because it protects file transfers inside an SSH session.
- Even when SFTP traffic is captured, the packet data is not useful to an attacker without the encryption keys.
- Organisations should use SFTP or another encrypted transfer method for sensitive data movement.

Conclusion

- The lab successfully demonstrated the security difference between FTP and SFTP.
- FTP allowed the username, password and transferred file content to be read from the Wireshark capture.
- SFTP transferred the same file through SSH on port 22, where the capture showed encrypted SSHv2 packets.
- FTP should not be used for sensitive transfers unless protected by a separate secure tunnel.
- SFTP is the safer option because it protects credentials and prevents captured traffic from exposing file content.

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References

- Group 6 FTP vs SFTP Security Demonstration Report.
- Wireshark packet capture evidence from the FTP and SFTP lab.
- Wireshark documentation for packet capture and protocol filtering.
- OpenSSH/SFTP documentation for secure file transfer over SSH.
- vsftpd documentation for FTP server setup and transfer testing.

THANK YOU